

Topics Under HTML Form and Tables

Forms

- HTML Form Structure and Method (POST, GET)
- Input Types: Text, Email, Number, Date, Time, Password, select>option
- Labels, IDs, name and required, readonly
- Checkbox, Radio Buttons, and Select (Single/Multi)
- Submit and **Reset Buttons** and Form Submission

Tables

- HTML Table Structure: Rows, Columns, and Cells
- Semantic Tables with thead, tbody, and tfoot
- Table Attributes: **colspan**, **rowspan**, classes, and IDs
- Special Table Styling : `<colgroup>/<col>`
- Size Chart Example: E-Commerce Style Table with thead and tbody

Important Element and Attributes of Forms

What is Labels?

An element that used to define a **label** for an `<input>` (or other form controls) is the `<label>` tag. It improves **accessibility** and **usability** by associating descriptive text with a form control, allowing users to click the label to focus or activate the input.

What is ID?

ID is a selector that is used to select unique element from the Document (HTML)

1. ID can be used to create Links (Eg. **#hero-section**)
2. ID can also be used to apply styling (In CSS)
3. **In Forms: Id is used to create associated text using `<label>` Element with `for=""` attribute that links input's **id** attribute**

Two ways to create labels in form input field ?

1. Explicit (We've done this)
2. Implicit

What is **name** attribute in HTML?

The **name** in an <input> element is an **attribute** that specifies a name for the input field.

- It is mainly used to **identify form data when a form is submitted** and ~~to reference the element in JavaScript.~~
- When a form is submitted, only inputs with a **name** attribute will have their values sent to the server
- The **name** becomes the key in the key-value pair sent in the request.

What is **readonly** attributes?

Note: **readonly** and **required** attributes are **boolean attributes**

- Makes the input **non-editable** by the user, but still **focusable** and **submittable** with the form.

What is **required** attributes?

- Marks the input as **mandatory** for form submission and doesn't allow to submit the form until required field is provided by client

What is **value** attributes?

- The **value** in an <input> tag specifies the **initial or current value** of the input element, and its meaning depends on the input type.

Types of Button in Forms ?

Submit Button: Sends the form data to the server by validating form fields and redirect to action Page (Default Behaviours)

Reset Button: Resets all form fields to their **initial values**

Types of Methods in Forms?

GET : Form data is appended to the URL as query parameters (e.g., ?name=John&age=25).

Data is visible in the browser's address bar.

POST: Form data is sent **in the request body**, not in the URL, data is not visible in address bar

What is **action** attributes in form?

The action attribute in an HTML <form> element specifies **where to redirect** and **send the form-data when the form is submitted**

Tables element in HTML ?

The **<table>** element is used to represent **tabular data** — information arranged in rows and columns .

A table is made up of several related elements:

Emp Code	Emp Name	Job Title	Salary(LPA)
911	Zoya Shaikh	Developer	3.6
912	Lisa Dev	Tester	2.8
913	Deepak Jadeja	Developer	5.2
914	Aiti Yewale	Digital Marketing	4
915	Simran Rao	HR	5.6

<table>	Defines the table container.
<tr>	Defines a table row (where all <td> are inserted).
<th>	Defines a header cell <i>(bold and centered by default).</i>
<td>	Defines a standard data cell <i>(Actual data is passed here).</i>
<caption>	Adds a title or description for the table.
<thead>	Groups header rows. (All <th> are wrapped under this)
<tbody>	Groups body rows.
<tfoot>	Groups footer rows.

-----STYLING COLUMNS-----

<colgroup> / <col> The <colgroup> tag in HTML is used to group columns within a table and apply styles to them collectively.

Example:

```
<colgroup>
  <col span="" style="background-color: lightblue" />
  <col style="background-color: lightgreen" />
  <col style="background-color: red" />
</colgroup>
```

Special Attributes in Table

colspan: Specifies how many columns a cell **<td>** should span.

```
<td colspan="2">Content</td>
```

rowspan : Specifies how many rows a cell **<td>** should span.

```
<td rowspan="3">Content</td>
```

Table Styling

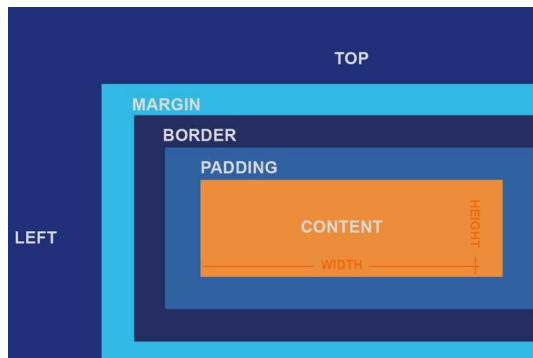
Let Understand what is BOX MODEL

Every Element in the HTML have **these four Properties** that can be modified based on the need!

Components of the Box Model

1. **Content**: This is the innermost part of the box where the **actual content**, such as **text** or **images**, is displayed.
2. **Padding**: This is the space between the content and the border. It is transparent and can be used to create space inside the element.
3. **Border**: This surrounds the padding and content. It can be styled with different colors, widths, and patterns.
4. **Margin**: This is the outermost layer that creates space between the element and other elements on the page. It is also transparent.

Demonstration with Example:



Apply styling to Table

```
td{  
  border: 2px solid rgb(247, 241, 241);  
  padding: 10px;  
}  
table{  
  border-collapse: collapse;  
}
```